RESP-100 · WEEK 1 · VISUAL STUDY GUIDE

Cardiac Anatomy & Physiology

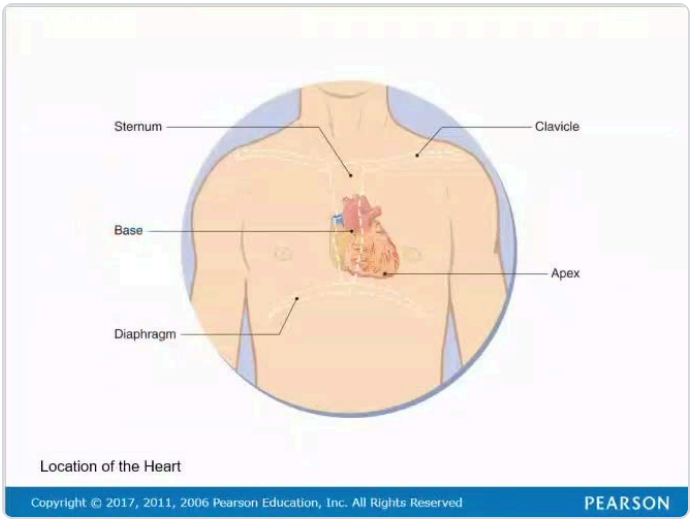
Walraven, *Basic Arrhythmias* 8th ed. · Appendix A · Lecture: Michael V Hyland (22 min)

Where the heart sits, its walls and valves, the path blood takes, how it pumps "from the top down", and which arteries feed which wall.

1 · Location | 2 · Layers of the wall | 3 · Blood flow | 4 · Valves | 5 · Systole, diastole & cardiac output | 6 · Coronary circulation & MI | 7 · Surfaces of the heart | Cheat sheet | Quiz

1 Where the heart sits

Big idea: the heart sits in the middle of the chest, slightly to the **left**, and **anterior** (toward the front), right under the sternum.



Slide: Location of the Heart. Base at the top, apex pointing down and to the left, resting on the diaphragm.

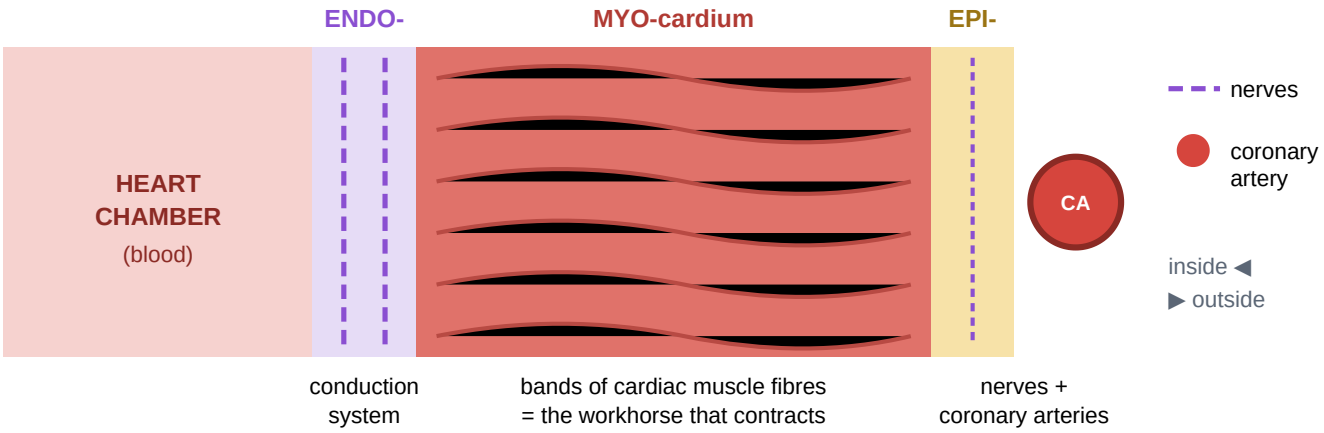
- **Middle** of the chest, slightly **left**
- **Anterior**: closer to the front than the back
- Right **under the sternum** (breastbone)
- **Base** = top (where the great vessels attach); **apex** = bottom tip, points left

WHY IT MATTERS FOR EKG

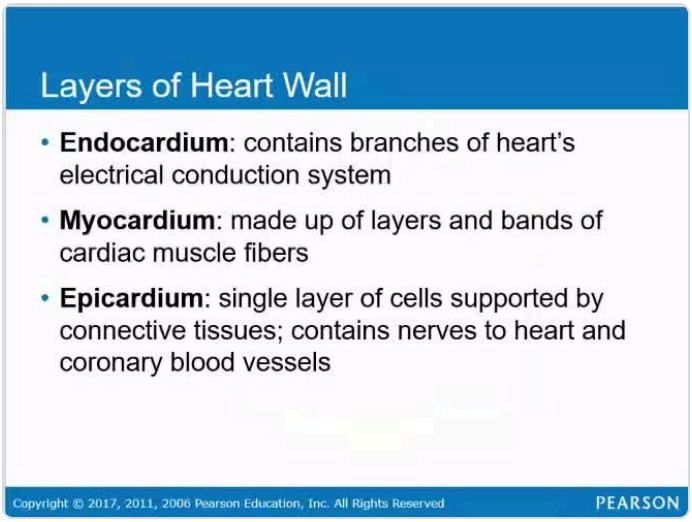
Electrodes for a monitor or a 12-lead must be placed correctly **around the heart's surface**, or the electrical reading won't be accurate.

2 Layers of the heart wall

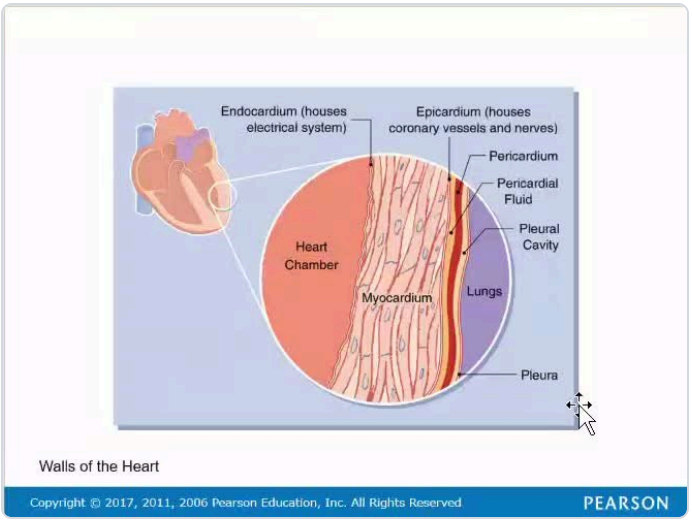
Big idea: three layers, inside to outside: **endo**cardium (wiring) → **myo**cardium (muscle, the workhorse) → **epi**cardium (nerves + coronary arteries).



My diagram: a slice through the heart wall from the chamber (left) to the outside (right).



Slide: Layers of Heart Wall



Slide: Walls of the Heart (pericardium, lungs and pleura sit outside)

MEMORY TRICK

Endo = inside (like *endo*scope). **Epi** = on top (like *epi*dermis). **Myo** = muscle (like *myo*sitis). Wiring inside, muscle in the middle, pipes on the outside.

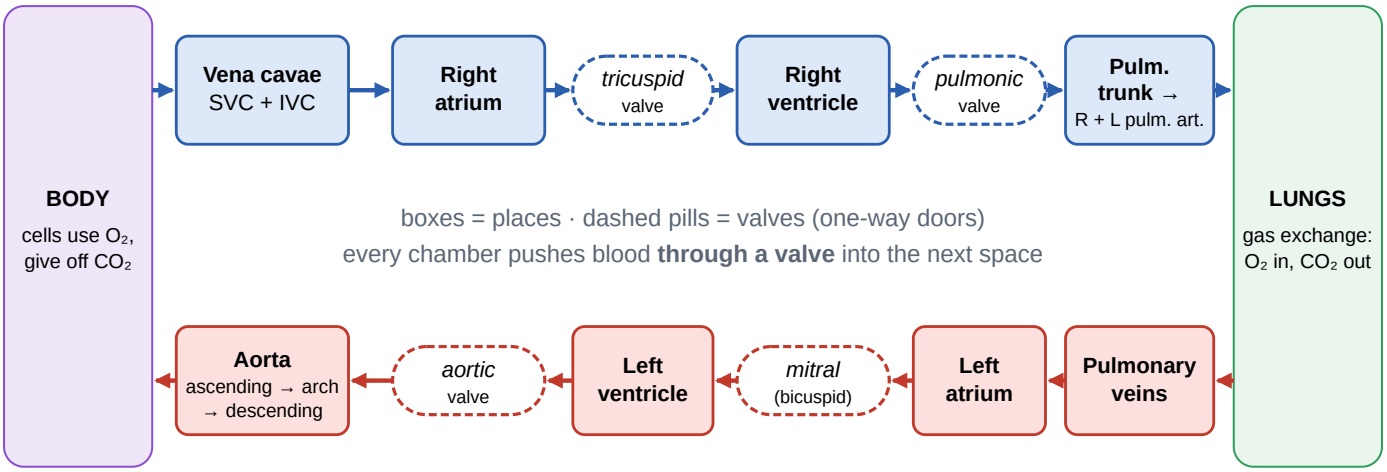
CLINICAL LINK

The coronary arteries in the epicardium come off the base of the aorta. A blockage there (coronary artery disease) cuts blood flow to the muscle cells, which can cause a **myocardial infarction** (heart attack).

3 Blood flow through the heart

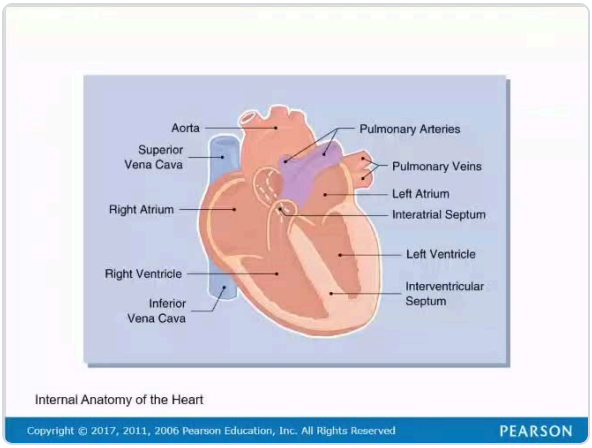
Big idea: learn the route **in order**. Right side = deoxygenated, **low pressure** (to the lungs). Left side = oxygenated, **high pressure** (to the whole body).

RIGHT HEART · deoxygenated (blue) · LOW pressure

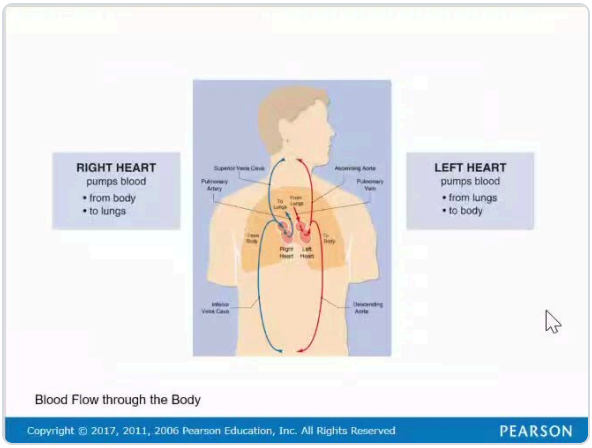


LEFT HEART · oxygenated (red) · HIGH pressure

My diagram: the full loop. Read the blue row left → right, then the red row right → left.



Slide: Internal Anatomy of the Heart. Note the interatrial and interventricular septa.



Slide: Blood Flow through the Body. Right heart: from body → to lungs. Left heart: from lungs → to body.

WHAT THE INSTRUCTOR STRESSED · 0:03:13

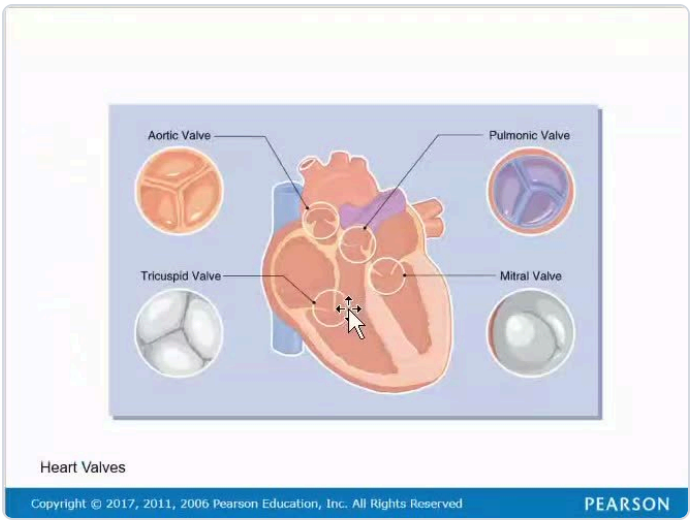
"Be organised" about blood flow: when someone has an arrhythmia or MI, knowing the normal flow helps you understand how the heart beats *abnormally*.

MEMORY TRICK

"Try before you buy": **Tri**cuspid comes before **Bi**cuspid (mitral). Right side first, then left. And **LAB**: **L**eft **A**V valve = **B**icuspid.

4 Heart valves

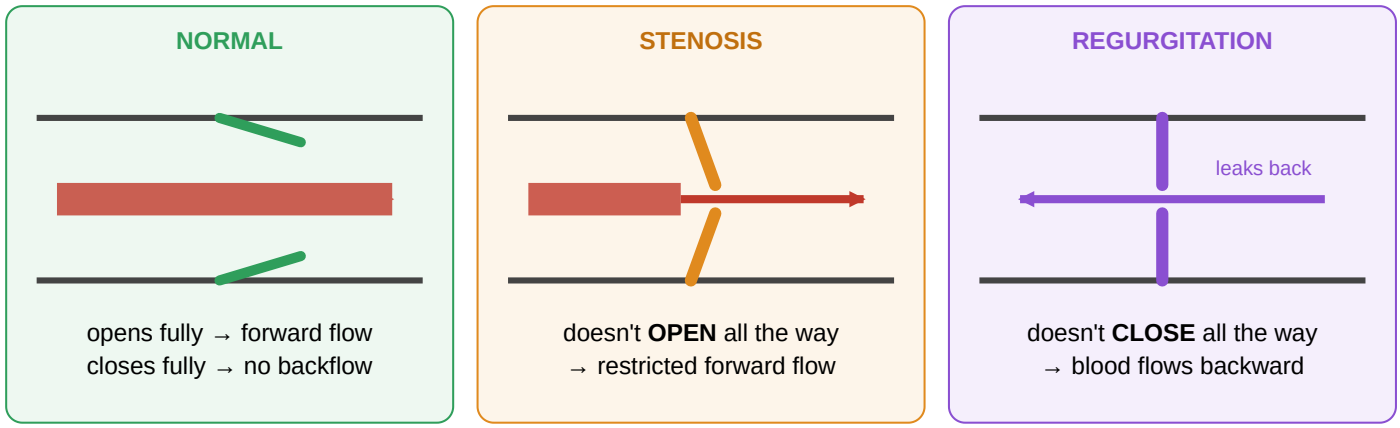
Big idea: 4 one-way valves. **AV valves** sit between atria and ventricles; **semilunar valves** sit at the base of the two big arteries.



Slide: Heart Valves (diagram)

Type	Right side	Left side
AV (atrio-ventricular) between upper & lower chambers	Tricuspid	Mitral (bicuspid)
Semilunar at the base of the arteries	Pulmonic (pulmonary semilunar) → pulmonary artery	Aortic (aortic semilunar) → aorta

Two ways a valve can fail



My diagram: blood flows left → right. Stenosis = a door stuck half-shut; regurgitation = a door that won't latch.

WHY AN EKG PERSON CARES · 0:10:10

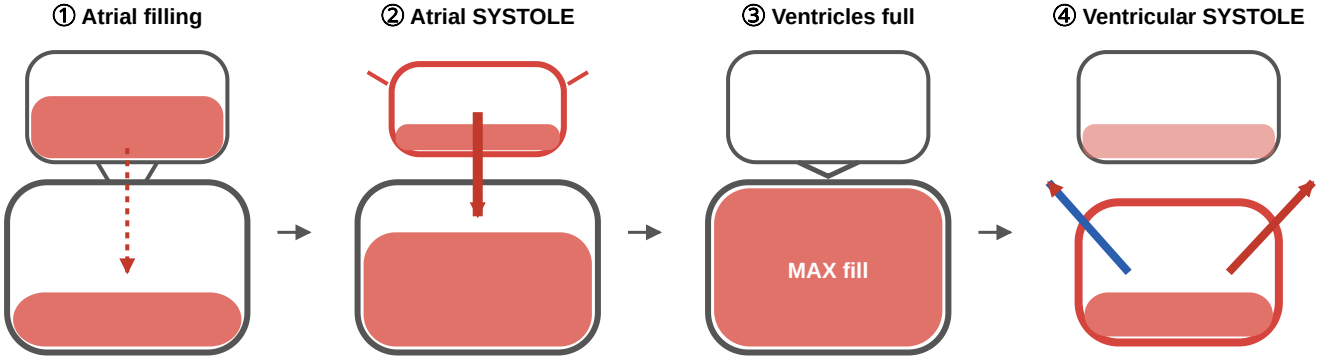
Valve problems can cause an **electrical variance** visible on a 12-lead EKG, and if untreated for a long time they can damage the right or left ventricle's function.

MEMORY TRICK

Stenosis = **S**tuck (won't open). **R**egurgitation = **R**everse (leaks back). **Semilunar** = half-moon shaped cusps guarding the two arteries.

5 Systole, diastole & cardiac output

Big idea: the heart works best when it beats **from the top down**: atria fill → atria contract → ventricles are maximally filled → ventricles contract. A fuller ventricle ejects more blood (bigger stroke volume).



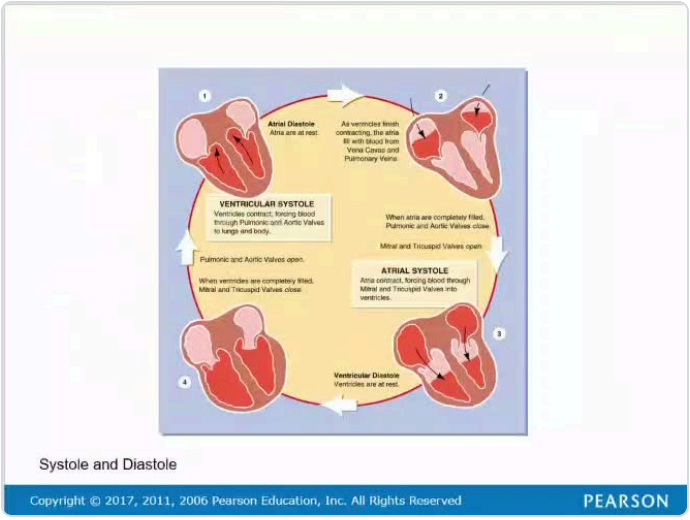
atria fill (IVC/SVC, pulm. veins); AV valves slightly open: some trickles

atria contract and push their blood **DOWN** into ventricles

maximised ventricular filling; AV valves close

ventricles contract: blood → pulm. artery + aorta (stroke volume)

My diagram: one beat "from the top down" (atria on top, ventricles below; red = blood). Atria refill during ④.

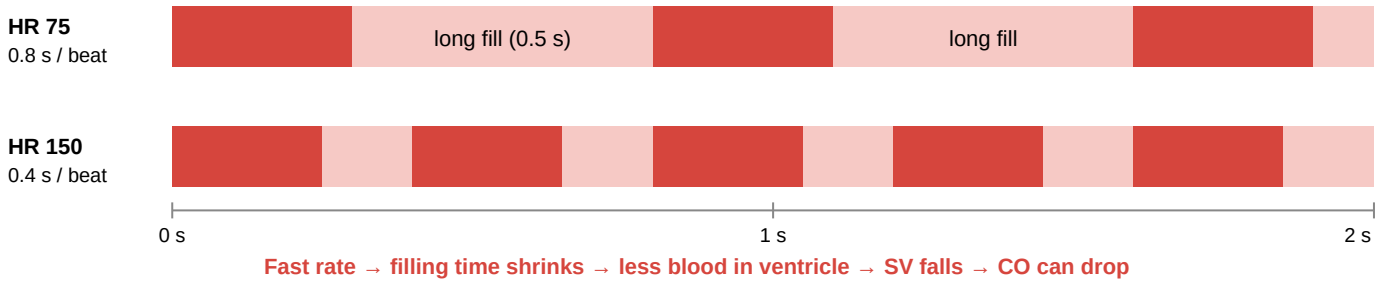


Slide: Systole and Diastole. The instructor started at ② (atrial filling/diastole), then ③, ④, and ① ventricular systole.

Word	Means the same as…
Diastole	relaxation · **filling** · **repolarization** (cells resetting to accept the next stimulus)
Systole	**contraction** · **depolarization**
Atrial systole	atrial contraction = atrial depolarization
Ventricular diastole	ventricles relaxed and filling

Cardiac output = Stroke volume × Heart rate

Same 2 seconds, two heart rates (dark = systole/ejecting, light = diastole/filling)



Fast rate → filling time shrinks → less blood in ventricle → SV falls → CO can drop

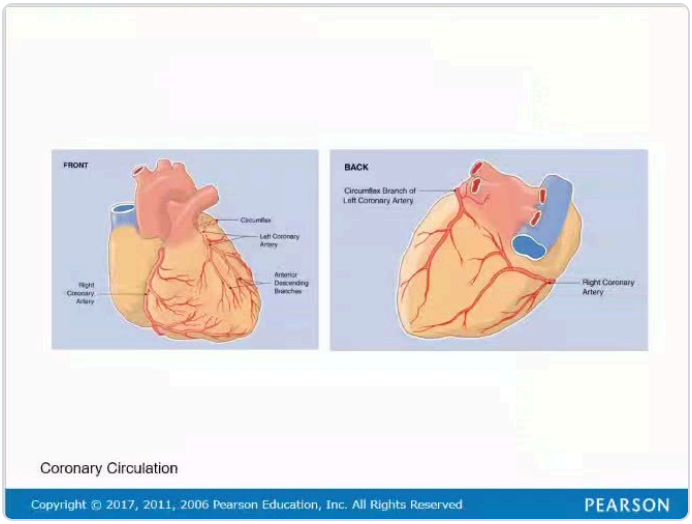
My diagram (approximate timings): speeding up mainly steals filling (diastole) time.

WHAT THE INSTRUCTOR STRESSED · 0:13:37

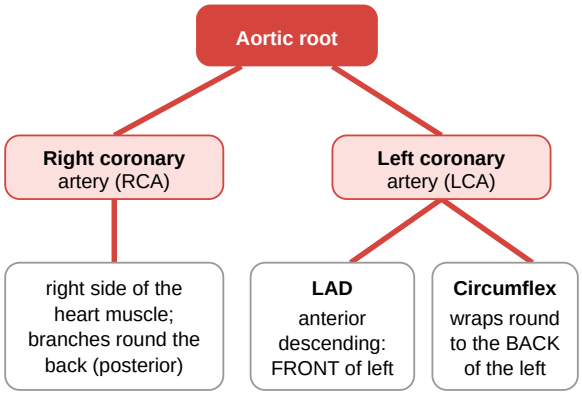
CO = SV × HR. Cardiac output is **best supported by stroke volume, not heart rate**. "The faster the heart contracts, the less time it has to fill with blood." These terms (systole/diastole) are reused all semester.

6 Coronary circulation & heart attack

Big idea: the coronary arteries branch off the root of the aorta and run on the **outside** of the heart to feed the heart muscle itself. Block one, and the muscle it feeds becomes ischemic → injured → dies (MI).

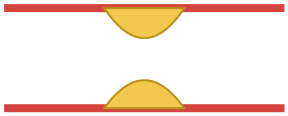


Slide: Coronary Circulation (front and back views)



My diagram: the coronary "family tree"

What a blockage does



plaque narrows artery

ISCHEMIA
not enough O_2 for cell metabolism

INJURY
blockage lasts longer

INFARCTION
cells die = myocardial infarction

The patient pathway the instructor described

Symptoms
SOB, chest pain, lightheaded, jaw / arm / leg pain, radiating back pain

Oxygen + 12-lead EKG
12 electrical views; abnormal lead → which surface / artery

Cath lab
cardiac cath: dye injected to find the blockage

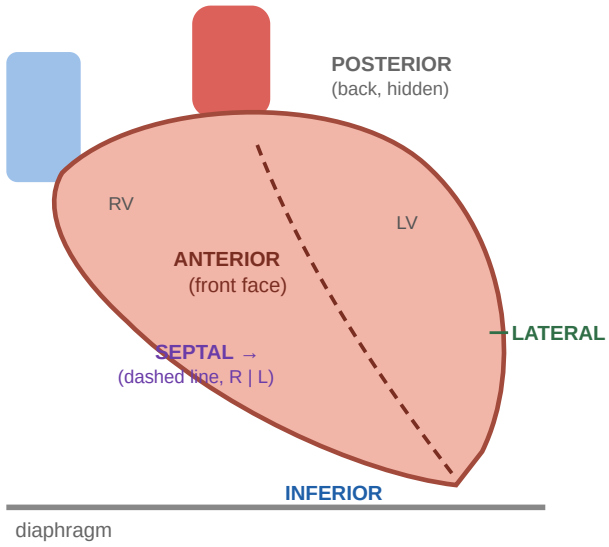
Fix it
coronary artery stent
or, if very bad, OR: open-heart surgery

WHAT THE INSTRUCTOR STRESSED · 0:17:31

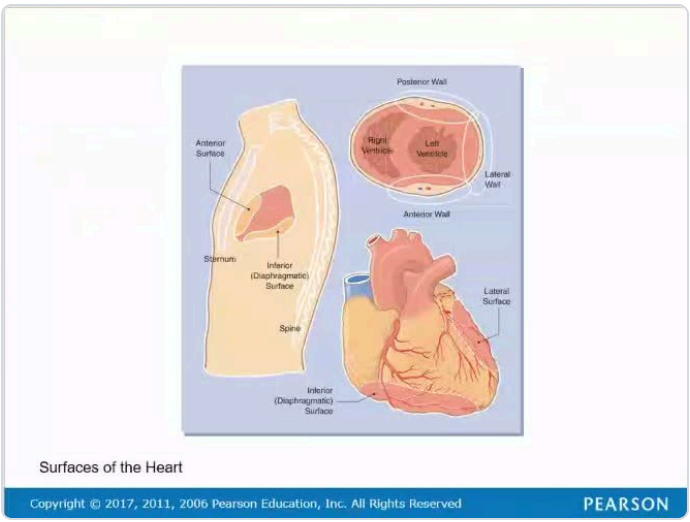
A blockage **high up** in the right coronary artery cuts flow to **all** the muscle cells it feeds downstream. The 12-lead tells us which **surface** is affected, which points to the culprit artery.

7 Surfaces (walls) of the heart

Big idea: five walls to name: **anterior, inferior, lateral, posterior, septal**. Each lead on a 12-lead "looks at" a surface, so an MI can be named by its wall.



My simplified front view (patient's left = your right). Dashed line = septum.



Slide: Surfaces of the Heart (side view and cross-section)

Wall	Where	Usual artery
Anterior	front of the left side	**LCA → LAD** (instructor's example)
Inferior	bottom, sits on the diaphragm	usually RCA (textbook)
Lateral	left side of the left ventricle	circumflex (textbook)
Posterior	back of the heart	RCA / circumflex branches around the back
Septal	band of muscle between right and left	LAD branches (textbook)

TIP

You don't need to read 12-leads yet. For now: **lead with abnormality → surface → artery**. You'll use this chain again in the 12-lead chapter.

★ One-page cheat sheet

Location & walls	
Location	middle of chest, slightly left, anterior, under the sternum
Endocardium	inner layer: conduction system
Myocardium	middle: cardiac muscle, contracts ("workhorse")
Epicardium	outer: nerves + coronary arteries

Valves	
AV valves	tricuspid (R), mitral/bicuspid (L)
Semilunar	pulmonic (R), aortic (L)
Stenosis	won't open fully → less forward flow
Regurgitation	won't close fully → backflow

Pumping words	
Systole	contraction = depolarization
Diastole	relaxation = filling = repolarization
Stroke volume	blood ejected per beat
Cardiac output	**SV × HR**; best supported by SV
Fast HR	less fill time → SV ↓ → CO ↓

Blood flow in one line

SVC/IVC → **RA** → *tricuspid* → **RV** → *pulmonic* → pulmonary trunk → R/L pulmonary arteries → **lungs** → pulmonary veins → **LA** → *mitral* → **LV** → *aortic* → aorta (ascending, arch, descending) → **body** → back to vena cavae

Pressure

Right = low pressure, deoxygenated, to lungs.
Left = high pressure, oxygenated, to body.

Top-down beat

atrial filling → atrial systole → max ventricular filling → ventricular systole

Coronaries & MI

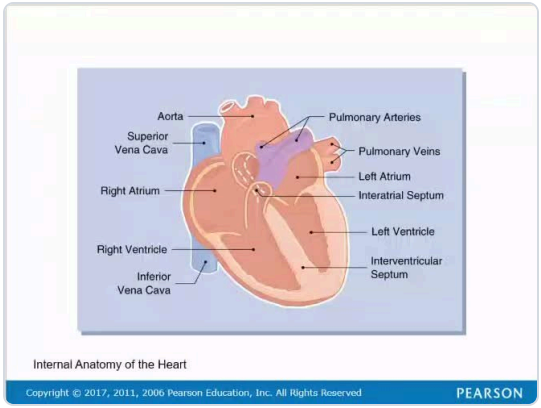
Off the aortic root. RCA: right side + posterior. LCA → LAD (anterior) + circumflex (back).
Blockage: **ischemia** → **injury** → **infarction**.
Workup: O_2 + 12-lead → cath lab → stent or open-heart surgery.

5 surfaces

Anterior · Inferior · Lateral · Posterior · Septal.
Mnemonic: "**A**ll **I** **L**ove **P**umps **S**trongly".

? Self-check quiz

1. Describe where the heart sits in the chest (3 details). Why does its position matter for EKG?
2. Which layer of the heart wall contains the electrical conduction system? Which contains the coronary arteries?
3. Put these in order of blood flow, starting from the vena cavae: left ventricle, pulmonic valve, right atrium, mitral valve, lungs, tricuspid valve, aorta, right ventricle, left atrium, aortic valve, pulmonary veins.
4. Name the two AV valves and the two semilunar valves. What is another name for the mitral valve?
5. A valve that doesn't open all the way is called _____; one that doesn't close all the way is called _____.
6. Why is the left side of the heart a high-pressure system and the right side a low-pressure one?
7. Define systole and diastole, giving the electrical word that goes with each.
8. Write the cardiac output formula. Why does a very fast heart rate lower cardiac output?
9. What does "the heart beats from the top down" mean, and why does it maximise stroke volume?
10. A clot blocks the right coronary artery high up. Name the 3 stages the muscle cells go through if it isn't relieved.
11. An EKG abnormality shows up in leads that view the **anterior** wall. Which artery is most likely involved?
12. On this slide, trace a drop of blood from the superior vena cava to the aorta. The valves are not labelled: name each valve you pass through. Which wall separates the two ventricles?



13. List the five surfaces (walls) of the heart. Which one rests on the diaphragm?

Answers

1. Middle of the chest, slightly left, anterior (under the sternum). Electrodes must be placed correctly around the heart's surface for an accurate electrical reading.
2. Endocardium = conduction system. Epicardium = nerves + coronary arteries.
3. Right atrium → tricuspid → right ventricle → pulmonic valve → lungs → pulmonary veins → left atrium → mitral → left ventricle → aortic valve → aorta.
4. AV: tricuspid (R), mitral (L). Semilunar: pulmonic, aortic. Mitral = bicuspid.
5. Stenosis (stenotic); regurgitation.

6. The left must pump blood all the way out to every cell, including hands and feet; pumping through the lungs needs little pressure.
7. Systole = contraction = depolarization. Diastole = relaxation/filling = repolarization.
8. CO = SV × HR. Faster rate → less time to fill → less blood in the chamber → stroke volume falls → CO can drop.
9. Atria fill and contract first, pushing all their blood into the ventricles before the ventricles contract; a fuller ventricle ejects more blood.
10. Ischemia → injury → infarction (cell death = myocardial infarction).
11. Left coronary artery, specifically the left anterior descending (LAD).
12. SVC → right atrium → *tricuspid* → right ventricle → *pulmonic* → pulmonary arteries → lungs → pulmonary veins → left atrium → *mitral* → left ventricle → *aortic* → aorta. The interventricular septum.
13. Anterior, inferior, lateral, posterior, septal. The inferior (diaphragmatic) wall.

Based on the Week 1 lecture "Appendix A: Cardiac Anatomy and Physiology". Items marked "textbook" are standard facts not stated in the lecture.